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RNA	-4.2 Bn		- First self-replicating genetic molecule in the RNA world i.e. Copied its complete genetic code when it split - -4.35Bn - Acts as both information carrier and catalyst (ribozymes) - -4.2Bn - Templatized the replication, and allowed room for variations , instead of exact copies - 4.2Bn	- Chemical conditions on early Earth favored stable, self-copying molecules - Environmental cycling: Wet-dry or hot-cold cycles likely promoted strand separation and copying, favoring an RNA-based replication system	- No longer capable of direct enzymatic activity like proteins (though ribozymes retained some function). - Extremely unstable and prone to degradation, which led to DNA eventually replacing it as the main genetic material.					
Prokaryotic Cells	-3.9 Bn	RNA	- First cell membranes formed as phospholipid bilayers enclosing self-replicating RNA - Cellular compartmentalization: a lipid membrane covering the RNA, to maintain an internal homeostasis , protecting replicators and concentrating reactants - By this time, RNA-directed protein synthesis may have evolved, so the first cells contained RNA plus encoded proteins - Primitive metabolism: Initially absorbed organic molecules from environment, but soon developed glycolysis/fermentation to generate ATP (all cells use ATP for energy) -Metabolic pathways: e.g. glycolysis to break down organics and synthesize ATP - Asexual reproduction by binary fission (inherited by all prokaryotes).- Protocell division: mechanism to segregate copies of RNA and contents into daughter cells (early cell cycle). - Had flagella for movement	- Resource depletion: Prebiotic organic "soup" was limited, driving evolution of internal energy generation (metabolism). - Isolation/protection: Enclosing replicating molecules in a membrane sheltered them from the harsh outside (UV, dilution), increasing survival	- Lost the ability to freely exchange genetic material like pre-cellular RNA life forms. - No longer had the flexibility of the RNA world's rapid molecular changes, making them more stable but less adaptable at the genetic level.					
Asgard Archaea	-3.5 Bn	Prokaryotic Cells	- Ether-linked membrane chemistry for stability in extreme conditions (e.g. thermophiles' membranes remain intact at high temperatures). - Novel metabolic pathways: e.g. methanogenesis to exploit $H_2 + CO_2$ reactions for energy – a new way to generate fuel in oxygen-free environments. - Extreme tolerance: enzymes and proteins adapted to high temperature or high salt, expanding life's range to previously uninhabitable niches.	- Selection favored organisms that could survive high heat and chemical extremes. - Nutrient scarcity: Different energy sources (hydrogen, sulfur, etc.) were utilized as organic nutrients were scarce; this drove Archaea to diverse chemistries. - Competition with bacteria: Archaea occupied niche habitats (e.g. hot vents, hypersaline pools) where their adaptations gave them an edge, avoiding direct competition and oxygen (which was absent or later toxic).							
Bacteria - Anaerobic	-3.5 Bn	Prokaryotic Cells	- Prokaryotic cells (no nucleus) with peptidoglycan cell walls (rigid support unique to bacteria). - Peptidoglycan cell wall: a tough polymer giving cells shape and protection – an innovation that distinguished Bacteria and allowed surviving varying osmotic conditions - Anaerobic metabolism: lived in oxygen-free conditions. Developed fermentation (e.g. glycolysis yielding ATP without O_2) and anaerobic respiration using alternative electron acceptors (e.g. sulfate or iron). - Anoxygenic photosynthesis: Some bacteria by ~3.5 Bn evolved primitive photosynthesis that did not produce oxygen (used H_2S or Fe^{2+} as electron donors). This exploited a proton gradient to make ATP – a mechanism adopted by virtually all later life-Photosynthetic pigments: (bacteriochlorophyll) to capture light energy (though anoxygenic), initiating light-powered autotrophy on Earth - Diversity of forms: spherical, rod, and spiral bacteria; some developed flagella for motility to seek nutrients	- No oxygen present: Metabolisms had to work without O_2, so bacteria exploited abundant chemicals; e.g. using sulfate or CO_2 as electron acceptors in respiration since the atmosphere was reducing. - Energy and carbon needs: As free organic molecules were used up, pressure favored bacteria that could make their own food (autotrophy). Sunlight was a vast energy source, driving evolution of photosynthesis despite the complexity of that pathway. - Rapid growth advantage: High mutation and reproduction rates in bacteria (vs. slower archaea) let them adapt quickly and fill myriad niches, diversifying their metabolic capabilities under competitive pressure.	- Lost ability to tolerate oxygen (oxygen was toxic to many early bacteria). - Became highly specialized in fermentation pathways rather than more efficient aerobic respiration.					
DNA	-3.5 Bn	RNA								
Cyanobacteria	-3 Bn	Bacteria - Anaerobic	- Oxygen release (Water-splitting): First organisms to produce free oxygen – a revolutionary innovation that transformed Earth's atmosphere. This required the photosystem II enzyme complex to extract electrons from water. - Stromatolite reef-building: By forming extensive microbial mats that trapped sediments, cyanobacteria pioneered large-scale biological structures (the first "reefs"), altering sediment and nutrient dynamics. - Protective pigments & enzymes: Evolved UV-screening compounds and antioxidant enzymes to survive their own oxygen output (which can form harmful reactive oxygen species). These innovations were new in an increasingly oxygenated world.	- Resource availability: Earlier anoxygenic photosynthesizers were limited by scarce H_2S/Fe^{2+}.	- Lost reliance on pre-existing organic molecules for energy, fully committing to photosynthesis (could no longer survive in the deep ocean without light). - Became dependent on sunlight, making them vulnerable to environmental shifts affecting light availability.					

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Aerobic Bacteria	-2.5Bn	Bacteria - Anaerobic	- Aerobic respiration: A new metabolic pathway unlocking much more energy from food. This innovation – oxidative phosphorylation – gave aerobes a huge advantage in energy yield, fueling more active lifestyles and larger cell sizes. - Reactive oxygen defenses: Enzymatic systems to neutralize peroxides and radicals (a novelty since earlier life didn't face oxygen). This was crucial for exploiting O_2 safely and is found in virtually all aerobes today. - Metabolic flexibility: Many aerobic bacteria became facultative – able to switch between aerobic and anaerobic modes. This flexibility was an innovation maximizing survival in fluctuating O_2 conditions (absent in strictly anaerobic ancestors).	- Oxygen "crisis": The buildup of O_2 (poisonous to anaerobes) exerted immense pressure – most anaerobic bacteria either perished or retreated to anoxic niches during the Great Oxidation Event. A few survivors evolved to utilize oxygen instead of being killed by it, turning a toxin into a resource. - Energy demands: As biosphere productivity increased (thanks to photosynthesis), competition for organic fuels intensified. Aerobes, with higher ATP output, could outcompete anaerobes for resources or inhabit new niches (e.g. consuming oxygenated organic detritus) that anaerobes couldn't tolerate. - Global change: Oxygenation of environments created new habitats (e.g. oxic surface oceans, soils). Bacteria that could exploit these oxic zones spread widely, driving evolutionary radiation. The presence of aerobic bacteria also set the stage for symbiotic relationships (leading to mitochondria in eukaryotes).	- Lost tolerance to anaerobic conditions; oxygen became necessary for survival. - Became reliant on oxidative metabolism, making them more efficient but requiring constant oxygen availability.					
Eukaryote with Mitochondria	-2.1 Bn	Aerobic Bacteria, Archaea	- Endosymbiosis (mitochondria): A revolutionary innovation – one cell living inside another in a mutualistic union. The engulfed bacterium provided abundant ATP, while the host provided nutrients and safety. This set the template for organelles. - Nucleus & organelles: First appearance of a nucleus (separating DNA from cytoplasm) and other organelles. This allowed spatial separation of transcription and translation, more complex gene regulation (splicing, mRNA processing), and storage of much more DNA. - Aerobic respiration internally: Mitochondria allowed highly efficient ATP production within the cell, supporting a larger cell size and genome. - Sexual reproduction: Likely present almost from the start – evidence suggests the earliest eukaryotes had meiosis and syngamy. They could recombine genes via sexual cycles, increasing genetic diversity	- Rising oxygen levels: By ~2.1 Bn, O_2 was sufficiently available; an anaerobic archaeal host benefited from teaming up with an aerobic bacterium to survive in an oxidizing world (the symbiotic bacterium consumed O_2 and provided energy). This partnership was favored in an O_2-rich environment where alone the host might struggle. - Energy and complexity needs: As cells grew larger and genomes expanded, a powerhouse was needed – mitochondria fulfilled the high ATP demand, permitting evolution of large, complex cells. Cells with mitochondria could out-compete those without by sustaining more active metabolism and genome maintenance.	- Lost the ability to fully survive without endosymbiotic mitochondria (became dependent on them for energy). - Lost rapid prokaryotic reproduction speeds due to the complexity of cell division.					
Eukaryote with Mito & Chloroplast	-2.1 Bn	Eukaryote with Mitochondria, Cyanobacteria	- Primary endosymbiosis (plastid): A second symbiotic breakthrough – a heterotrophic eukaryote permanently incorporated a photosynthetic prokaryote. This innovation of chloroplasts enabled eukaryotes to become autotrophic, a capability all earlier eukaryotes lacked. - Complete autotrophy: The cell could produce its own organic matter and oxygen. This was new among eukaryotes – it no longer relied on eating other organisms, fundamentally altering the flow of energy in ecosystems (eukaryotes could now act as primary producers). - Organelle integration: The host evolved novel cellular machinery to manage the chloroplast (protein targeting signals, regulation of chloroplast division, gene transfer to nucleus). This represented a higher level of cellular complexity – a cell with multiple endosymbiont-derived organelles functioning in concert.	- Nutrient advantage: In an ecosystem where competition for fixed carbon was fierce, housing a photosynthetic symbiont meant an endless food supply from sunlight. Selection strongly favored the host–symbiont union because the host gained nutrition without hunting, and the symbiont gained protection and access to inorganic nutrients (CO_2, minerals). - Environmental opportunity: By ~1.5–1.8 Bn years ago, oceans had moderate oxygen and ample light in surface waters. They could egulf cyano living in shallow waters and spread to new habitats.	- Lost the ability to consume organic food sources directly (became reliant on photosynthesis for survival). - Reduced motility compared to heterotrophic eukaryotes, as they no longer needed to hunt for food.					
Red Algae	-1.5 Bn	Eukaryote with Mito & Chloroplast		- Ecological competition: In crowded marine microbial mats, becoming multicellular gave red algae an edge – they could rise above surfaces to access sunlight and weren't as easily grazed upon or outcompeted for space. The increase in oxygen and nutrient availability around ~1.2–1.0 Ga allowed larger, more complex algae to thrive. - Genetic variation: Environments in Proterozoic oceans were changing (nutrient flux, perhaps occasional global glaciations). Sex provided a way to rapidly generate variation, helping algae adapt. The success of sexual red algae likely pressured other protists to adopt sexual cycles as well.	- Lost the ability to efficiently use all wavelen - Lost the flagella present in many single-cell	- True multicellularity in eukaryotes: Red algae were the earliest organisms known to develop a complex multicellular body with specialized cells (a major leap from unicellular ancestors). This allowed larger size and shape (e.g. leaf-like thalli) for better light capture and stability on substrates. - Sexual reproduction with gametes: Introduced gamete differentiation (eggs and spermatia) and fusion in eukaryotes, greatly increasing genetic diversity. The oldest evidence of sex comes from red algal fossils - this innovation accelerated eukaryotic evolution by enabling recombination. - Calcium carbonate deposition (later innovation): Some red algae later evolved the ability to calcify their cell walls (forming coralline algae), helping build reefs. This wasn't present in the earliest reds, but the group eventually pioneered this among algae, influencing marine ecosystems.				year
Fungi	-1.2 Bn	Eukaryote with Mitochondria								paremt
Green Algae	-1.0 Bn	Red Algae								

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Sponges	-800 Mn	Eukaryote with Mitochondria								
Fungi on Land	-730 Mn	Fungi								
Comb Jellies - Ctenophores	-700 Mn	Eukaryote with Mitochondria	Updated to a Complex Nervous System							
Bilateria	-600 Mn	Eukaryote with Mitochondria								
Land Green Algae	-600 Mn	Green Algae								
Corals, Sea Anemones, Jellyfish - Cnidarians	-580 Mn	Eukaryote with Mitochondria								
Earliest Protosome	-590 Mn	Bilateria								
Earliest Deuterosome	-570 Mn	Bilateria								
Ambulacaria - 1st Dueterosome Split	- 570 Mn	Earliest Deuterosome								
Chordate - 2nd Dueterosome	-541 Mn	Earliest Deuterosome								
Mollusks - Earliest Protosome Split	-540 Mn	Earliest Protosome								
Annelids - 2nd Protosome Split	-530Mn	Earliest Protosome								
Arthropods - earliest Trilobites	-520 Mn	Earliest Protosome								
Lancelets	-520 Mn	Chordate - 2nd Dueterosome								
Plants Colonize Land	-515 Mn	Land Green Algae								
Parasites (Some Arthropods evolve parasitic property)	-512Mn	**Started with Arthropods? or Only in Arthropods?**								
Earliest Cephalopods	-500Mn	Mollusks - Earliest Protosome Split								
Euthycarcinoids	-500Mn	Arthropods - earliest Trilobites								
Vertebrates - 1st one - Jawless fish	-500Mn	Chordate - 2nd Dueterosome								
Bony Fish	-420Mn	Jawless fish								
Scorpions	-420Mn	Arthropods - earliest Trilobites								
Plants take root	-410Mn	Land Plants								
Plants develop woody stems	-400Mn	Rooted Plants								
Cartilignous Fish	-400Mn	Jawless Fish								
Lobe Finned Fish	-400Mn	Bony Fish								
Trees	-385Mn	Rooted Plants								
Tiktaalik	-375Mn	Lobe Finned Fish								
Earliest Tetrapods	-365Mn	Tiktaalik								
Insects grow in size, variety & winged insects	-360Mn	Arthropods - earliest Trilobites								
Early Crab, Shrimps, etc	-360Mn	Arthropods - earliest Trilobites								
Ferns	-360Mn	Rooted Plants								
Amphibians	-360Mn	Earliest Tetrapods								
Amniotes	-320Mn	Earliest Tetrapods								
Earliest Saurapids -1st split in Amniotes	-320Mn	Amniotes								
Earliest Synapsids - 2nd split in Amniotes	-320Mn	Amniotes								

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Turtles <family name?> - 1st Split in Saurapids	-300Mn	Earliest Saurapids								
Reptiles - Saurapids - 2nd Split	-300Mn	Earliest Saurapids								
First Octopus	-296Mn	Earliest Cephalopods								
Crocodiles	-230Mn	Earliest Saurapids								
Dinosaurs	-230Mn	Earliest Saurapids								
Earliest Mammals	-225Mn	Earliest Synapsids								
Lizards & More Reptiles	-200Mn	Reptiles								
Monotromes	-166Mn	Earliest Mammals								
Marsupials	-160Mn	Earliest Mammals								
Pollinating Insects	-160Mn	Insects - Arthropods								
Birds	-150Mn	Dinosaurs								
Earliest Placental Mammals	-120Mn	Earliest Mammals								
Flowering Plants	-120Mn	Rooted Plants								
Tortoises	-100Mn	Turtles								
Snakes	-100Mn	Reptiles								
Grass	-70Mn	Flowering Plants								
Dinosaurs Extinction	-66Mn									
Aftrotheres - 1st Split in Placental Mammals	-65Mn	Earliest Placental Mammals								
Xenartheria - 2nd Placental Split	-65Mn	Earliest Placental Mammals								
Laurasiatheria	-65Mn	Earliest Placental Mammals								
Euarchantoglires	-65Mn	Earliest Placental Mammals								
Elephants	-60Mn	Aftrotheres								
Armadillos	-60Mn	Xenartheria								
Earliest Rodents	-60Mn	Euarchantoglires								
Rodents, Group1 - Rats, Mice, Squirrels, Beavers	-60Mn	Earliest Rodents								
Earliest Primates	-60Mn	Euarchantoglires								
Sloths, Anteaters	-55Mn	Xenartheria								
Earliest Even Toed Hoofed Mammals	-55Mn	Laurasiatheria								
Even Toed Hoofed Mammals, Group 1 - Deers, Pigs, Camels, Bovines	-55Mn	Earliest Even Toed Hoofed Mammals								
Bats	-55Mn	Laurasiatheria								
Shrews, Hedgehogs	-55Mn	Laurasiatheria								
Rodents - Group 2 - Rabbits, Hares	-55Mn	Earliest Rodents Euarchantoglires								
Earliest Odd Toed Hoofed Mammals	-55Mn	Laurasiatheria								
Odd Toed Hoofed Mammals, Group1 - Rhinos	-55Mn	Earliest Odd Toed Hoofed Mammals								
Even Toed Hoofed Mammals, Group 2 - Whales, Dolphins	-50Mn	Earliest Even Toed Hoofed Mammals								
Odd Toed Hoofed Mammals, Group 2 - Horses	-50Mn	Earliest Odd Toed Hoofed Mammals								
Lemurs	-50Mn	Earliest Primates								

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Carnivores - Wolfs, Wild Dogs, Bears, Cats, Seals, Weasles	-40Mn	Earliest Odd Toed Hoofed Mammals Laurasiatheria								
Monkeys, Apes	-40Mn	Earliest Primates								
Great Apes	-20Mn	Apes								
Hominims	-7Mn	Great Apes								
Homosapiens	-0.3Mn	Hominims								
Laurasiatheria										
Pangolins ~55 Ma — scaly mammals, sister group to Carnivora (the only mammals with true scales)										
From Euarchontogl	Rats To Humans									
Treeshrews (Scandentia) ~60 Ma — small squirrel-like mammals, actually closer to primates than to rodents										
Colugos (Dermoptera) ~60 Ma — the "flying lemurs" (they don't fly and aren't lemurs), gliding mammals that are the closest living relatives of primates										
Entry	**Date**	**Parent**								
Afrotheria	-65 Ma	Earliest Placental Mammals								
Xenarthria	-65 Ma	Earliest Placental Mammals								
Laurasiatheria	-65 Ma	Earliest Placental Mammals								
Euarchontoglires	-65 Ma	Earliest Placental Mammals								
Elephants, Manate	-60 Ma	Afrotheria								
Armadillos	-60 Ma	Xenarthria								
Earliest Rodents	-60 Ma	Euarchontoglires								
Rodents Grp 1 — F	-60 Ma	Earliest Rodents								
Lagomorphs — Ra	-55 Ma	Euarchontoglires								
Earliest Primates	-60 Ma	Euarchontoglires								
Treeshrews, Colug	-60 Ma	Euarchontoglires								
Sloths, Anteaters	-55 Ma	Xenarthria								
Earliest Even-Toed	-55 Ma	Laurasiatheria								
Even-Toed Grp 1 –	-55 Ma	Earliest Even-Toed								
Bats	-55 Ma	Laurasiatheria								
Shrews, Hedgehog	-55 Ma	Laurasiatheria								
Pangolins	-55 Ma	Laurasiatheria								
Earliest Odd-Toed I	-55 Ma	Laurasiatheria								
Odd-Toed Grp 1 —	-55 Ma	Earliest Odd-Toed								
Even-Toed Grp 2 –	-50 Ma	Earliest Even-Toed								
Odd-Toed Grp 2 —	-50 Ma	Earliest Odd-Toed								
Lemurs	-50 Ma	Earliest Primates								
Tarsiers	-55 Ma	Earliest Primates								
Carnivores — Wolv	-55 Ma	Laurasiatheria								
New World Monkey	-40 Ma	Earliest Primates								
Old World Monkeys	-25 Ma	Earliest Primates								
Gibbons (Lesser Ap	-20 Ma	Earliest Primates								
Great Apes — Orar	-14 Ma	Earliest Primates								
Gorillas	-10 Ma	Great Apes								
Hominins	-7 Ma	Great Apes								
Chimpanzees, Bon	-7 Ma	Great Apes								
Homo sapiens	-0.3 Ma	Hominins								